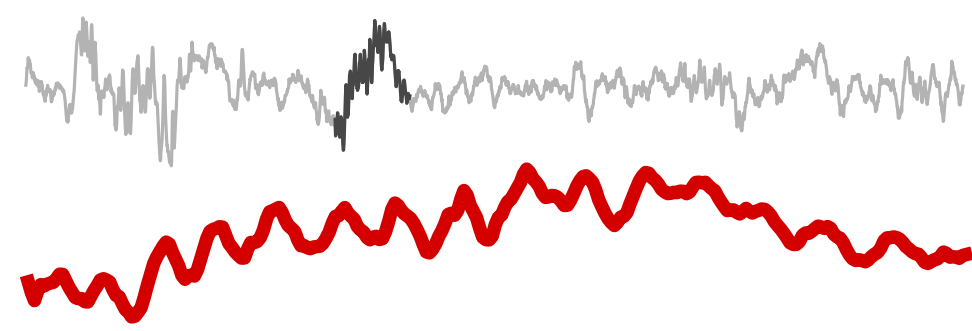


Sleep Spindles

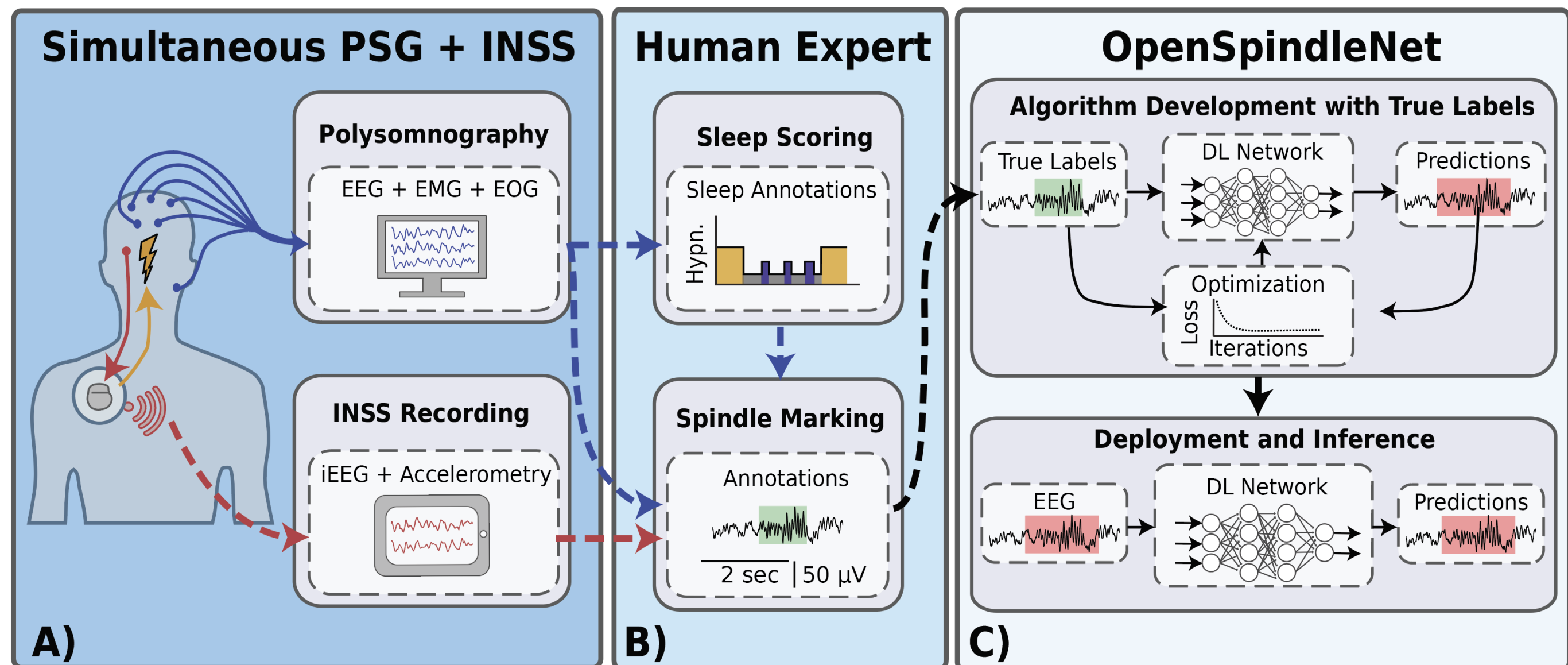


- memory consolidation
- sleep quality
- cognitive functions

Automatic detection

- brain health assessment
- sleep state physiology
- understanding disorders

Methodology



Datasets

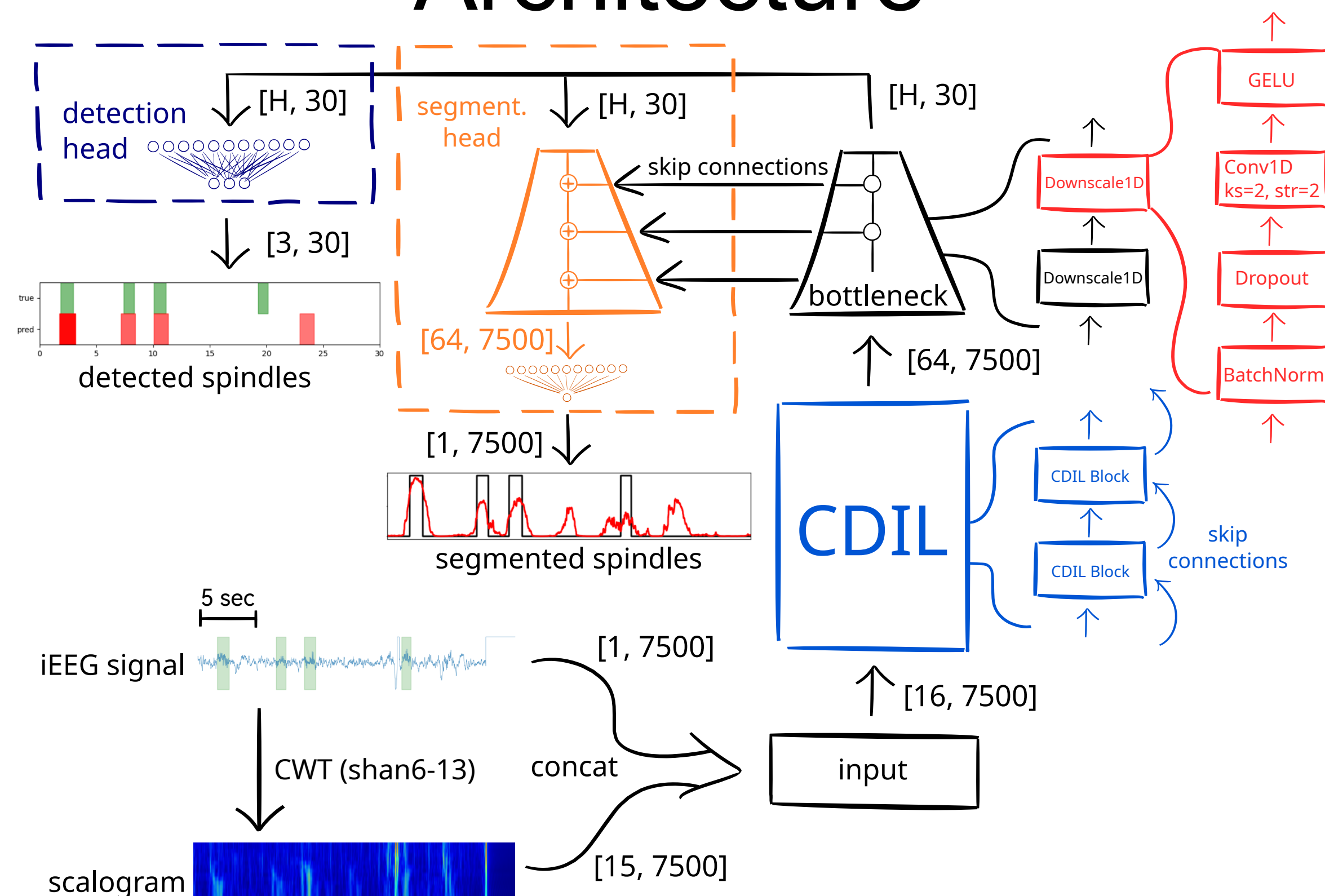
Metric	BNEL iEEG	DREAMS Scalp EEG
Subjects	4 Patients	8 Healthy Subjects
Total Duration	5.1 hours	2.0 hours
Annotated Spindles	1,030	578

Two distinct datasets: a public scalp EEG (DREAMS) and a private intracranial iEEG (BNEL) dataset.

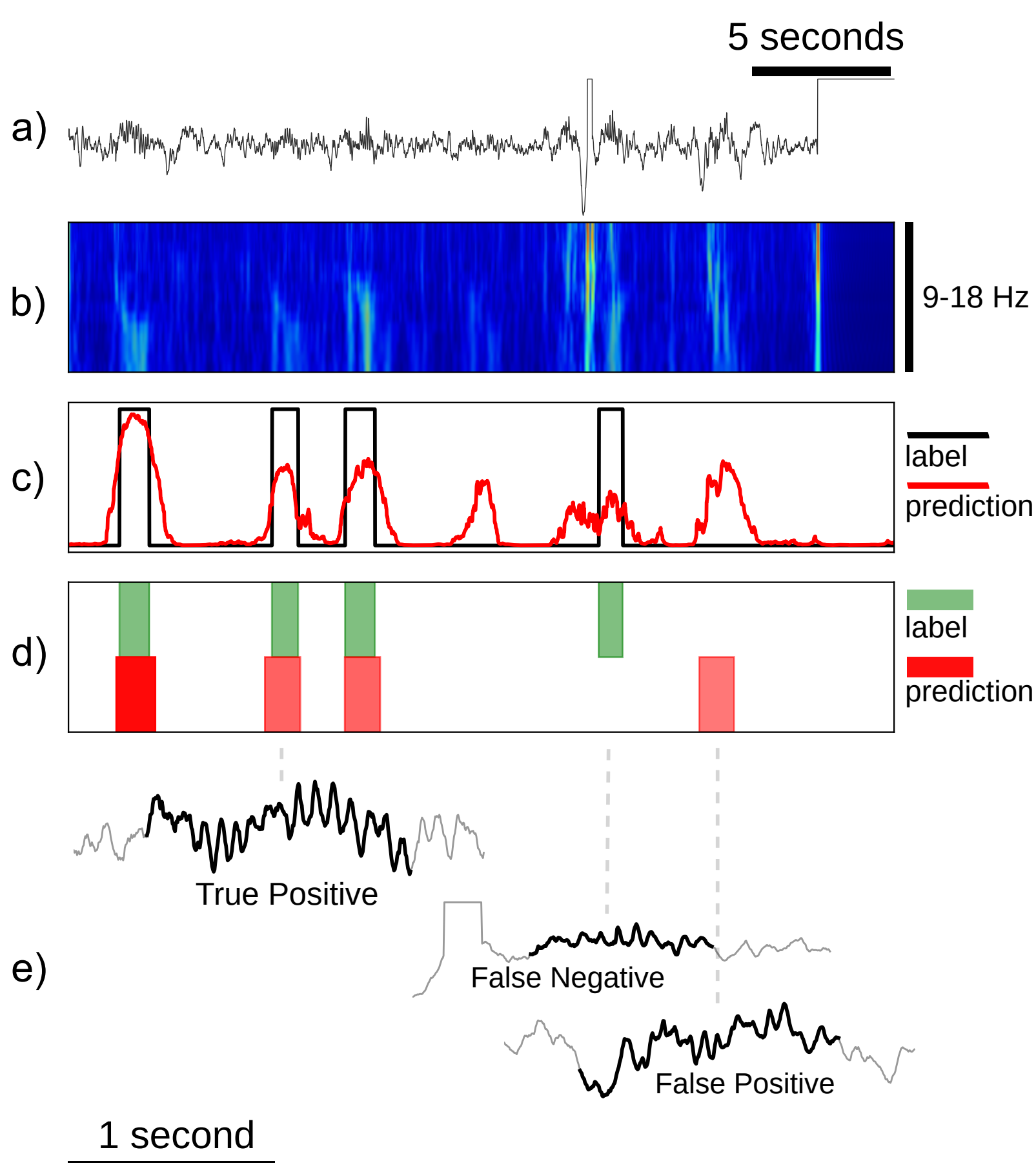
DREAMS: Recordings from 8 healthy subjects.

BNEL iEEG: Recordings from 4 epilepsy patients.

Architecture



Detections

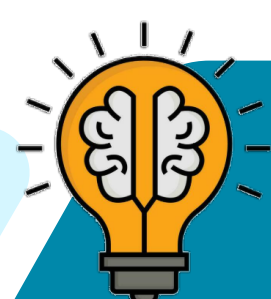


High Performance: Achieves a high F1-score of 0.67 on the complex intracranial EEG dataset, outperforming other methods.

Dual Output: Provides both discrete event markers (detection) and a continuous probability score (segmentation).

Accurate Identification: Effectively identifies true spindles while minimizing incorrect detections (false positives) and missed events (false negatives).

(a) The normalized, time-domain input signal. (b) The corresponding continuous wavelet transform (CWT) scalogram. Panels (a) and (b) together form the dual-channel input to the model. (c) The output from the model's segmentation head, where the black outlines denote ground truth and the red line shows the model's continuous prediction confidence. (d) The final discrete event predictions from the detection head (red blocks) compared against the ground truth labels (green blocks). (e) Enlarged examples from the raw signal, highlighting a true positive (a correctly identified spindle), a



Conclusion: We compared our method to various state-of-the-art methods on multiple datasets. Our approach provides a reliable and effective solution for spindle detection (F1=0.67), outperforming existing methods in key performance metrics.